

QUANTITATIVE TRADING WITH MACHINE LEARNING

Problem #43 | Method 2: Extended Model Comparison and Trading Evaluation

1. Problem Statement

The project investigates whether machine learning can extract a useful signal for future Volkswagen stock returns from historical market information. After an initial paper-inspired rolling evaluation produced weak aggregate out-of-sample results, a second experiment was conducted to compare alternative regression models and examine whether their predictions could support a trading strategy. This second experiment is an extension, not an exact reproduction of the reference paper.

2. Dataset Details

Method 2 uses **databaseFrankfurtComplete.csv**, a Frankfurt-market dataset containing approximately 3,809 rows and 1,292 columns from January 2005 to December 2019. The target stock is identified by ticker **VOW.F**. The dataset includes closing prices and numerous return-related features. The pipeline uses the closing-price series to construct a forward-return target and engineers additional momentum and rolling-volatility features over 5, 10, 20 and 60 observations.

3. Approach

Observations are sorted by date and divided chronologically: the first 60% are used for model fitting and the remaining 40% for evaluation. The experiment compares three regressors:

- **Ridge Regression:** linear regression with L2 regularisation to shrink coefficients.
- **Elastic Net:** combines L1 and L2 regularisation.
- **Extra Trees Regression:** an ensemble of randomized decision trees that can model nonlinear relationships.

For a horizon of h observations, the target is defined as $y(t) = \text{Close}(t+h) / \text{Close}(t) - 1$. Predictions are converted into trading signals using thresholds: sufficiently positive predictions indicate long positions, sufficiently negative predictions indicate short positions in long-short mode, and predictions near zero remain neutral. The backtest includes a simplified turnover cost and is compared with a buy-and-hold benchmark.

4. Implementation Overview

The Python implementation loads the CSV with pandas, inspects the data, sorts it chronologically, prepares features and the forward-return target, handles missing values using median imputation, and scales inputs for linear models where appropriate. It fits the three regressors, generates predictions, and calculates forecasting and trading metrics. Forecast metrics include mean absolute error (MAE), a zero-prediction baseline MAE and directional accuracy. Trading metrics include compounded return, approximate annualised Sharpe ratio, maximum drawdown and percentage of observations invested.

5. Results and Conclusions

In the reported chronological 60:40 experiment, Ridge achieved a strategy return of 81.71% and an approximate Sharpe ratio of 0.500, compared with a reported buy-and-hold return of 10.71% and Sharpe ratio of 0.201. Elastic Net and Extra Trees reported strategy returns of 43.48% and 47.97%, respectively. However, all three models had higher test MAE than the zero-prediction baseline (0.0671): Ridge 0.0683, Elastic Net 0.0684 and Extra Trees 0.0736. Therefore, the backtest results are exploratory and do not establish reliable predictive power or guaranteed profitability.

Limitations: The experiment uses a single chronological split; trading thresholds were selected using fit-period predictions, which may produce optimistic results. Feature availability at trade time, overlapping target horizons, transaction costs and execution assumptions require further validation. A stronger evaluation would use a separate validation period, freeze model and threshold choices, and report results on an untouched test period.